



JUNIOR DATA ENGINEERING PROGRAMME
GENERATION SINGAPORE (In partnership with Microsoft and Temasek Polytechnic)

FROM PUCK DROP TO VICTORY

NHL Win-Factor Analysis

Agenda

- 1 Business Problem & Objectives
- 2 Data Scope, Pipeline & Model
- 3 Data Preparation & Methodology
- 4 Findings & Dashboard
- 5 Validation & Challenges
- 6 Recommendations & Next Steps

The Business Problem

NHL game data is rich but fragmented — spread across many CSV files with different grains and data-quality issues, creating a risk of double counting, inconsistent calculations and misleading conclusions.

MAIN BUSINESS QUESTION

Which in-game performance factors are most strongly associated with winning an NHL game?

- How do winners differ from losers across scoring, special teams, physical play and goaltending?
- How does win rate change across low-to-high performance bands?
- Do relationships differ between regular season and playoffs, and across seasons?

Project Objectives

1

Design

Build a medallion ETL/ELT pipeline in Microsoft Fabric

2

Prepare

Explore, clean, and validate the NHL datasets

3

Model

Create a team-game fact table with reusable dimensions

4

Analyze

Build five analytical Delta tables and a four-page dashboard

5

Validate

Test data integrity, calculations and dashboard

6

Recommend

Translate the findings into appropriately qualified guidance

Data Source & Scope

NHL Game Data — Kaggle (Martin Ellis)

NHL data spanning 2000/01–2019/20, covering 19 played seasons

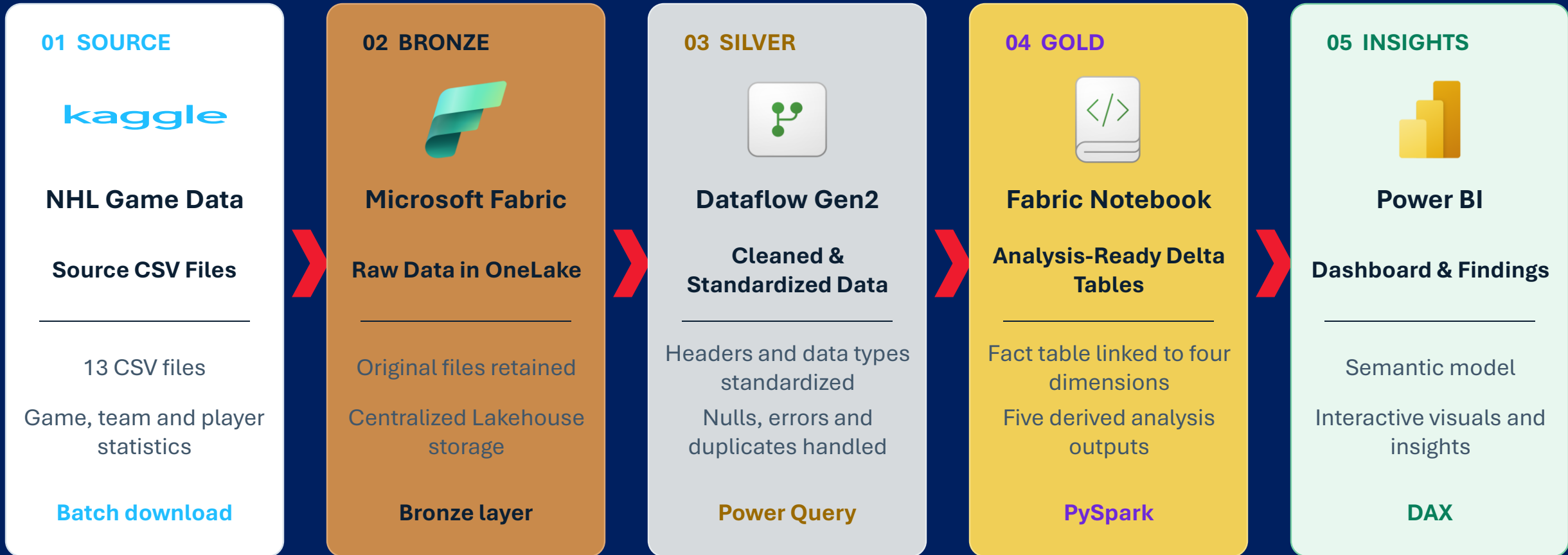
[kaggle-nhl-game-data](#)

IN SCOPE

- ✓ Team-game, skater and goalie statistics
- ✓ Regular-season and playoff comparisons
- ✓ Season-level performance trends
- ✓ Data-coverage rules for fair comparisons

Architecture: The Medallion Pipeline

NHL game data moves from raw files to validated analysis outputs and an interactive dashboard

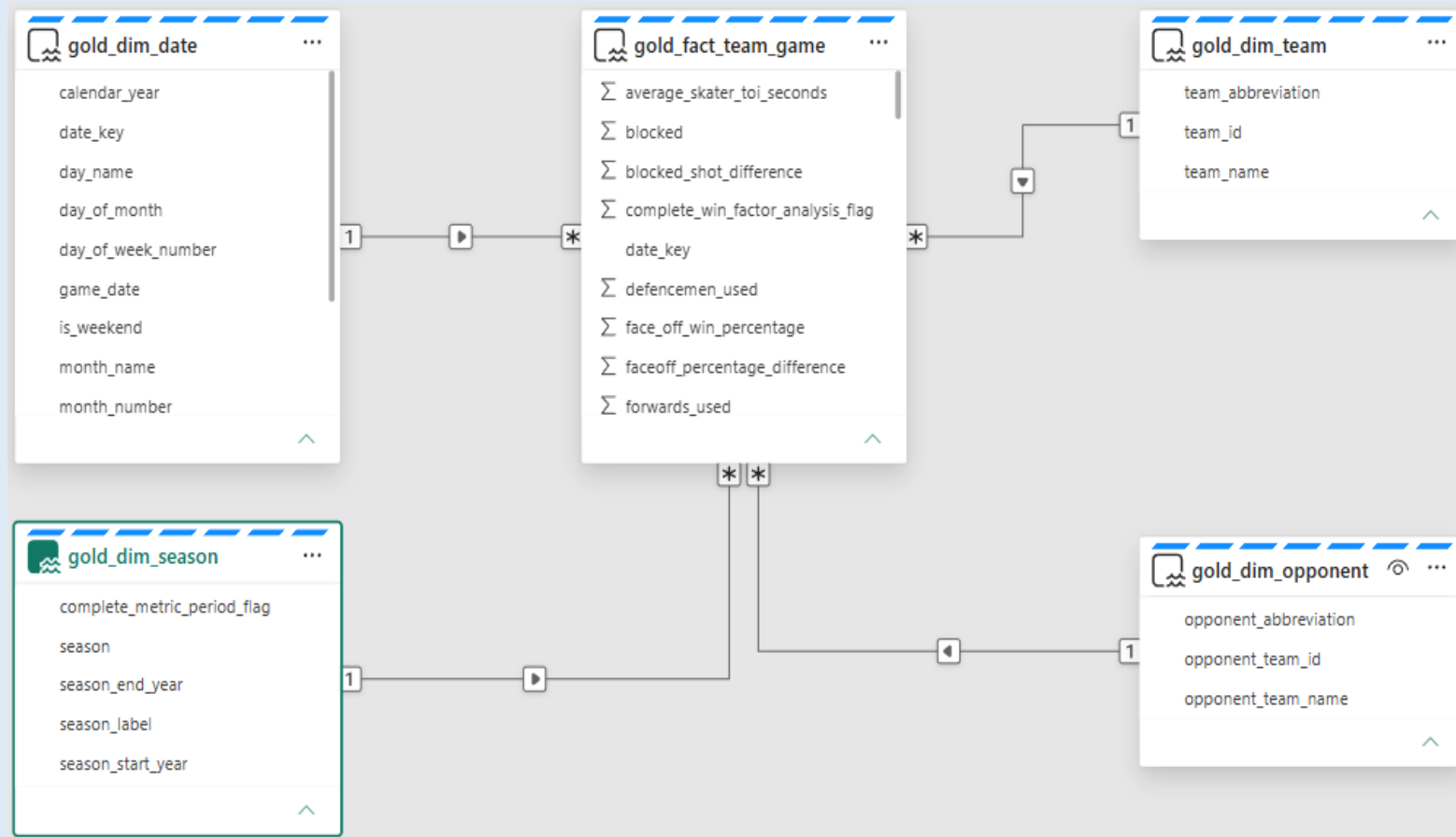


Kaggle CSVs → Fabric Lakehouse Bronze → Dataflow Gen2 Silver → PySpark Gold Delta → Power BI semantic model

Gold Data Model

The star schema keeps one row per game per team.

Five derived analysis tables were built from the same curated Gold dataset used to create the star schema.



Data
Tables
Model
Search
> gold_analysis_01_winner_loser_averages
> gold_analysis_02_win_factor_correlations
> gold_analysis_03_performance_bands
> gold_analysis_04_regular_vs_playoffs
> gold_analysis_05_season_trends
> gold_dim_date
> gold_dim_opponent
> gold_dim_season
> gold_dim_team
> gold_fact_team_game
> gold_nhl_win_factors

Dataset Selection & Preparation

All 13 Kaggle CSVs were landed in the Lakehouse; analysis used the subset needed for game, team, player and goalie context.

Dataset	Grain / Purpose	Key Quality Action
silver_game	1 row per game; 23,735 records; date, season, game type and final score	Headers repaired; data types converted; duplicate game_id values checked
silver_game_teams_stats	1 row per game per team; goals, shots, power play and penalties	Numeric columns profiled; conversion issues handled as nulls
silver_team_info	1 row per team; team lookup information	Headers and data types checked; team_id uniqueness validated
silver_game_skater_stats	1 row per game per skater (game_id + player_id)	Composite-key duplicates reviewed; numeric fields validated
silver_player_info	1 row per player; player lookup information	Duplicate records and data types checked; player IDs standardised
silver_game_goalie_stats	1 row per game per goalie (game_id + player_id)	Composite-key duplicates reviewed; numeric fields validated

Data Quality & Historical Completeness

2000/01 to 2001/02

Hits, giveaways, takeaways, blocked shots & faceoff win percentage missing

Excluded from physical-play comparison

2002/03 to 2009/10

faceoff win percentage; other physical metrics available

Excluded from the complete five-metric comparison

2010/11 onwards

All five physical / puck-management metrics complete

Used as the fair comparison window

Analysis rule:

Complete win-factor comparisons are restricted to 2010/11 onwards, when all selected metrics are available.

Transformation - Silver Layer

Dataflow Gen2 & Power Query

- Promoted and corrected column headers
- Assigned text, whole-number, decimal, date and datetime types
- Removed confirmed duplicate rows using the correct business key
- Converted parsing errors to null only after confirming they represented historically unavailable data, not corrupted values
- Removed unused linkage columns (e.g. venue_link) that did not support analysis
- Configured Lakehouse destinations and published/refreshed the Dataflow
- Reprofiled the transformed tables to confirm valid, error and empty-value percentages

Gold Analytical Outputs

1 Winner vs. Loser Averages

Compare average performance metrics for winning and losing teams

2 Win-Factor Correlations

Use Pearson correlation to rank associations with winning; correlation indicates association, not causation.

3 Performance Bands

Compare win rates across low-to-high performance quartiles

4 Regular Season vs. Playoffs

Compare factors by game type, noting the smaller playoff sample

5 Season Trends

Track whether relationships remain consistent across seasons

Key Finding: Shooting Efficiency Leads

$r = 0.58$

Shooting Percentage
Association with winning

Shooting percentage has the strongest measured association with winning among the factors analysed.

Goal-scorer count, point-contributor count and save percentage also show positive relationships. Home advantage exists but is small ($r = 0.09$). Hits, penalty minutes and giveaways show weak or near-zero relationships when considered individually.

Takeaway

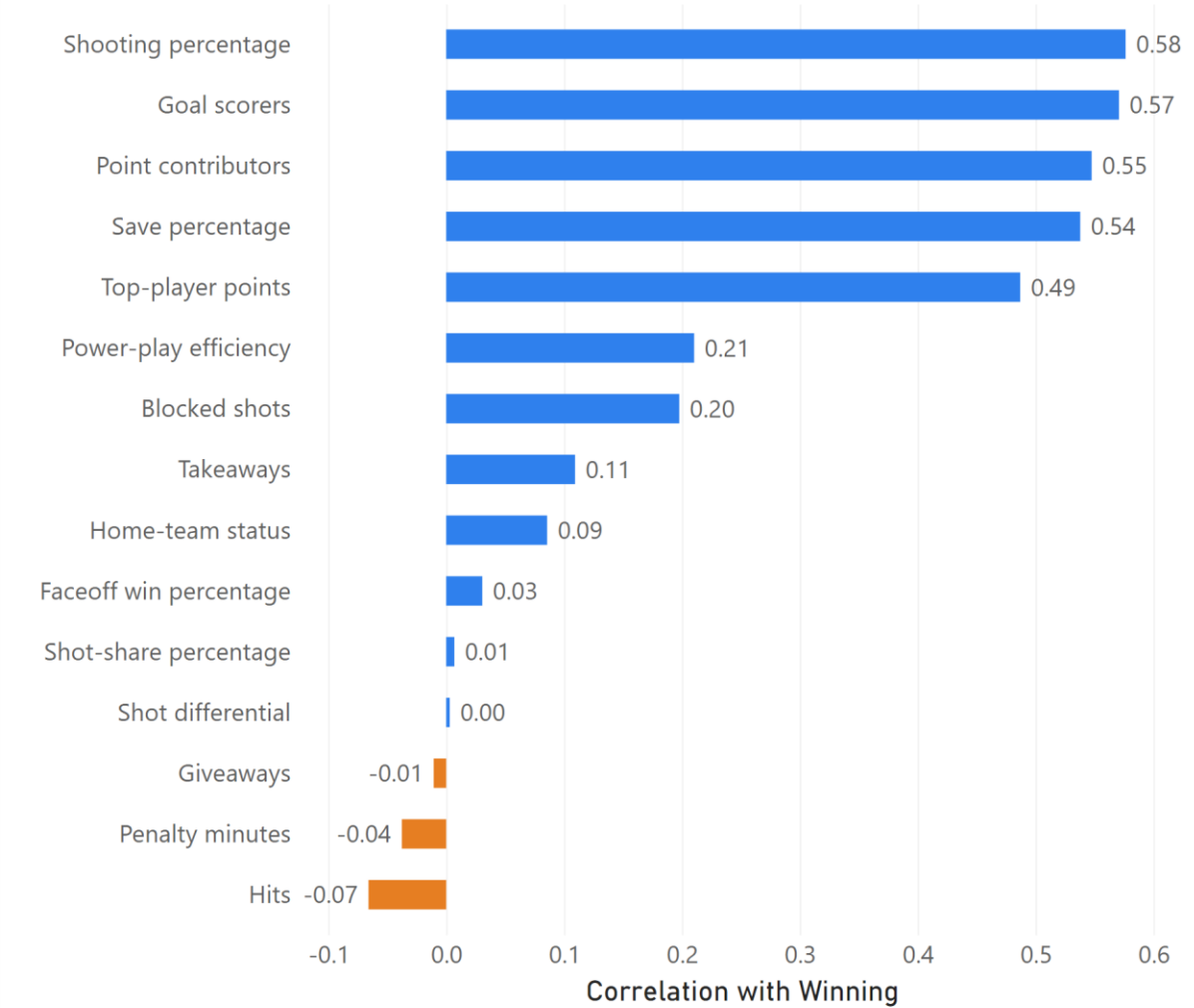
Shooting efficiency—not shot volume alone—shows the clearest measured association with winning.

Win-Factor Overview

Performance factors associated with winning NHL games

Analysis window: 2010/11–2019/20

Win Factors Ranked by Correlation



How Winners Differ from Losers

Metric	Winner Avg	Loser Avg	Winner-Loser Gap
Blocked shots	15.41	13.39	2.01
Faceoff win percentage	50.22	49.78	0.44
Giveaways	8.77	8.87	-0.10
Goal scorers	3.29	1.67	1.61
Hits	23.00	24.14	-1.14
Home-team share	0.54	0.46	0.09
Penalty minutes	9.73	10.32	-0.60
Point contributors	7.42	4.18	3.24
Power-play efficiency	23.94	13.47	10.46
Save percentage	94.00	88.03	5.97
Shooting percentage	12.72	6.01	6.71
Shot difference	0.03	-0.03	0.06
Shot share percentage	50.06	49.94	0.12
Takeaways	7.36	6.53	0.83
Top player points	2.17	1.27	0.90

Special-teams note:
Power-play and penalty-kill percentages mirror each other, so they are treated as one special-teams factor.

Key takeaway:
Shooting percentage has the strongest link to winning ($r = 0.58$), followed by goal scorers, point contributors, and save percentage. Other metrics show weak correlations. These are associations, not proof of causation.

Performance Bands

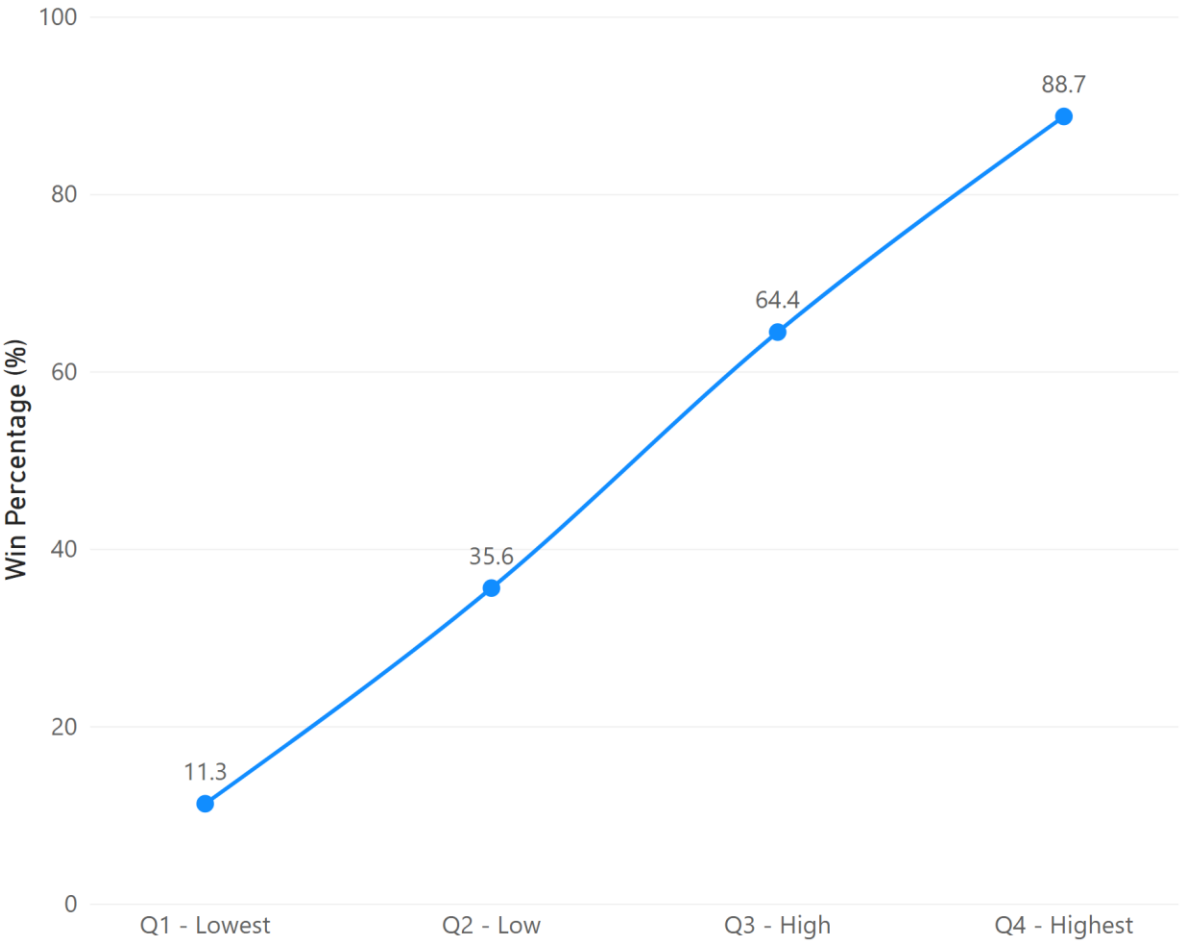
How win rate changes across different levels of team performance

Analysis window: 2010/11–2019/20

Select Win Factor

Shooting percentage

Win Percentage by Performance Band



Selected Factor Band Details

Band	Minimum	Maximum	Average	Team-Games	Win %
Q1 - Lowest	0.00	5.00	2.62	6323	11.26
Q2 - Low	5.00	8.70	6.86	6323	35.57
Q3 - High	8.70	13.04	10.71	6323	64.43
Q4 - Highest	13.04	45.45	17.26	6323	88.74

Lowest Band Win %

11.3

Highest Band Win %

88.7

How to read this page:

- A steep upward line means higher values of the selected factor are strongly associated with winning. A downward line indicates a negative relationship, while a nearly flat line indicates little separation in win percentage across the four performance bands.
- Quartiles contain approximately equal numbers of team-game records. Tied or rounded values may appear in adjacent bands

01 Win-Factor Overview

02 Performance Bands

03 Regular vs Playoffs

04 Season Trends

Regular Season vs Playoffs

How winning factors differ between regular-season and playoff games

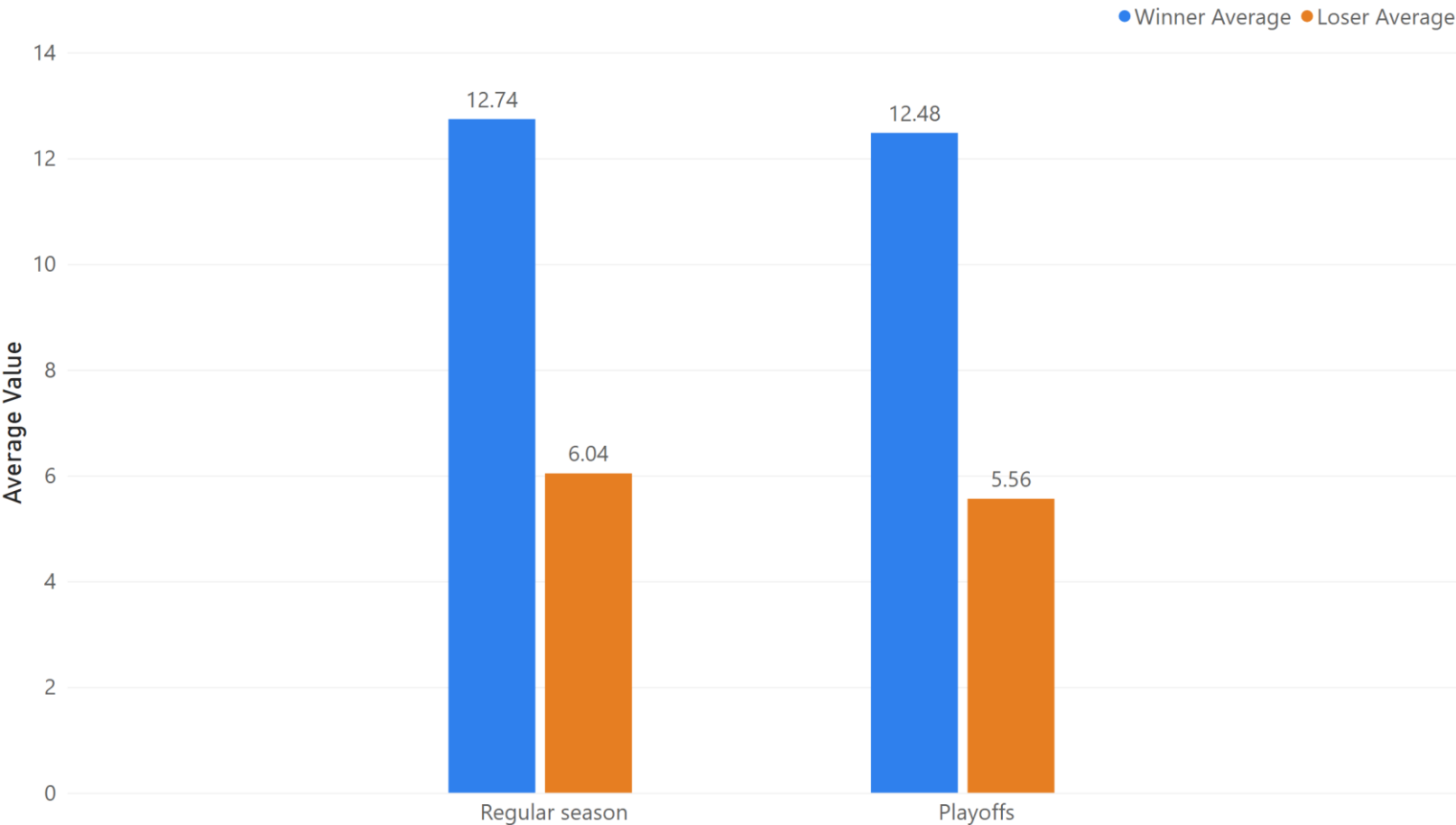
Analysis window: 2010/11–2019/20

Select Win Factor

Shooting percentage



Winner and Loser Averages by Game Stage



Regular-Season Winner–Loser Gap

6.70

Playoff Winner–Loser Gap

6.93

How to read this page:

A larger absolute winner–loser gap means the selected factor separates winners from losers more strongly within that game stage. A positive gap means winners recorded more; a negative gap means winners recorded less. Playoff results are based on fewer games, so they may fluctuate more than regular-season results.

Season Trends

How winner-loser performance gaps have changed across NHL seasons

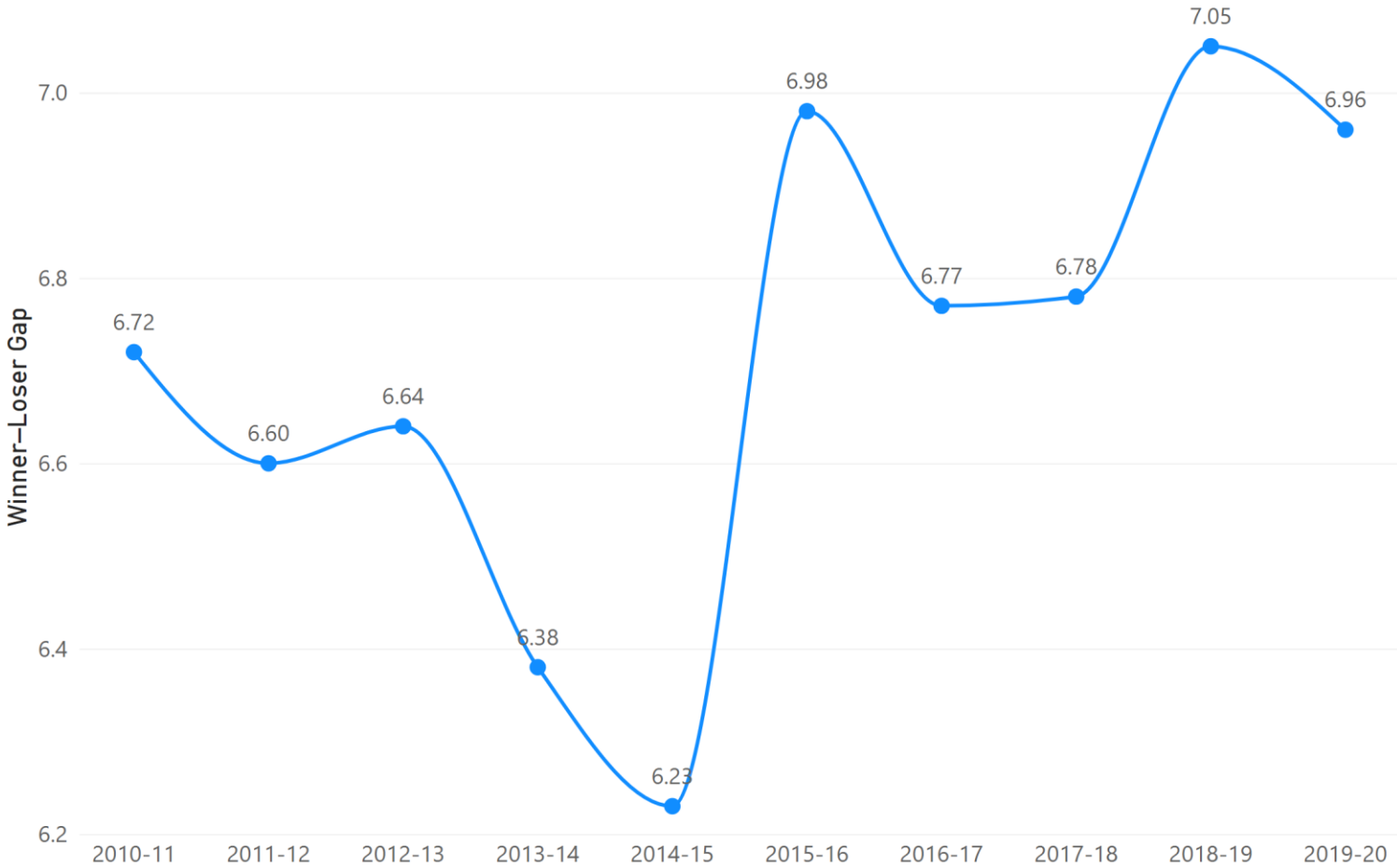
Analysis window: 2010/11–2019/20

Select Win Factor

Shooting percentage



Winner–Loser Gap by Season



Season Detail

Season	Winner Avg	Loser Avg	Winner–Loser Gap
2010-11	12.64	5.92	6.72
2011-12	12.53	5.93	6.60
2012-13	12.60	5.95	6.64
2013-14	12.39	6.00	6.38
2014-15	12.26	6.02	6.23
2015-16	12.77	5.79	6.98
2016-17	12.65	5.88	6.77
2017-18	12.87	6.09	6.78
2018-19	13.26	6.21	7.05
2019-20	13.21	6.25	6.96

Trend note:

The Y-axis automatically adjusts to the selected factor because the metrics use different units and scales. Compare the displayed values rather than judging the apparent steepness of the line. Trends cover 2010–11 to 2019–20, the period used for consistent win-factor coverage.

01 Win-Factor Overview

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Business Questions Answered

Business Question	Key Finding
Which factors are most strongly associated with winning?	Shooting percentage showed the strongest measured association with winning ($r = 0.58$).
How do winning teams differ from losing teams?	Winning teams generally showed higher shooting efficiency, broader player contribution and stronger save percentage.
How does win rate change across performance bands?	Win rate generally increased across stronger performance bands, especially for shooting percentage.
Do the relationships differ by game type or season?	The main patterns were broadly consistent, although playoff samples were smaller and results varied somewhat by season.

Results show descriptive associations and do not establish causation

Testing & Validation

PIPELINE VALIDATION

- ✓ All 13 source files present in Bronze
- ✓ No unexplained duplicates at intended grain
- ✓ Headers and schema repaired; data types corrected
- ✓ Two team rows per completed game; outcomes reconciled, including historical ties

DASHBOARD VALIDATION

- ✓ Slicer & cross-filter interaction verified
- ✓ Chronological season ordering
- ✓ Performance-band results reconciled: 11.26% (Q1) → 88.74% (Q4)
- ✓ Default metric (shooting %) restored on reset

Acceptance status: *The four-page dashboard passed data, calculation, interaction and visual-quality validation.*

Challenges & Resolutions

Challenge	Resolution
Incorrect CSV headers during ingestion	Corrected headers in Dataflow Gen2; reprofiled full dataset
Duplicates in game & player/goalie sources	Identified correct grain/key; removed duplicates & validated uniqueness
Ten team-stat columns contained valid, error and empty values	Reviewed patterns; converted genuine unavailable values to null
Historical statistics incomplete for older seasons	Added completeness rules; restricted physical-play comparisons to 2010/11 onwards
Multiple source grains risked row multiplication	Aggregated skater/goalie data to game-team level before joining
Two team-level Gold rows per game could cause double-counting	Used DISTINCTCOUNT(game_id) for game totals
Unplayed 0–0 placeholder games could distort results	Identified and excluded 14 unplayed fixtures from the Gold analytical population.

Recommendations

- 1 Use shooting percentage as a descriptive performance indicator, not evidence of a causal strategy
- 2 Track scoring contribution across lines and players
- 3 Review offensive and goaltending efficiency together
- 4 Treat home advantage as context, not strategy

Limitations & Next Steps

Limitations

- Observational analysis shows association, not causation
- Shooting percentage includes goals, which directly determine outcomes
- Missing older-season metrics restrict comparable analysis windows
- Playoff samples are smaller and represent qualifying teams only
- Pearson correlation captures linear association only

Next Steps

- Automate ingestion and create run-level data-quality logs
- Extend the analysis to game_plays.csv using partitioned processing
- Add opponent strength and score-state context
- Evaluate logistic-regression and tree-based models
- Extend the dimensions to player and game-stage attributes if reporting expands

THANK YOU